

Somasekhar Agoli

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Summary

Network Automation Engineer with **2.5 years of experience** building Python-based automation for network security infrastructure validation, reliability testing, and operational troubleshooting workflows. Proven track record of automating testing and validation across enterprise network security platforms (**FMC/FTD/cdFMC**) and **CI/CD pipelines**, cutting manual effort by up to **90%**.

Skilled in **Python**, **REST API automation**, **network automation**, and infrastructure-driven testing, with experience building reusable automation modules, maintainable validation frameworks, and repeatable troubleshooting workflows.

Leveraged **AI-assisted development workflows** (GitHub Copilot, Cursor, Windsurf) to accelerate automation development, debugging, and test-case generation.

Experience

Cisco Systems

Automation Engineer

Nov 2023 – Present

Bangalore, India

Project 1: High-Availability Validation Framework – Cisco Secure Firewall Management Center (FMC)

- Built a **pyATS**-based CI/CD automation framework to validate FMC High-Availability deployments, covering failover, state synchronization, policy replication, and configuration consistency across active/standby nodes.
- Automated 46+ parameterized HA scenarios spanning base, corner, negative, and failure-condition paths, with pre/post validation gates.
- Automated DNS/FQDN resolution validation for FMC HA pair configuration, ensuring ActiveStandby IP mappings resolve correctly across network zones before HA setup.

Project 2: Disaster Recovery Automation – Cloud-Delivered FMC (cdFMC)

- Engineered a DR automation framework for cdFMC covering snapshot lifecycle management, recovery orchestration, and repeatable post-recovery validation.
- Automated snapshot creation, rollback, configuration restoration, policy recovery, and device-registration workflows.
- Implemented post-recovery validation gates for network connectivity, IP reachability, HA synchronization, and event/log consistency across managed devices.
- Integrated scheduled and on-demand DR runs into CI/CD with network topology validation and service discovery checks.
- Improved DR validation efficiency by ~90% and reduced manual recovery effort by ~80%.

Project 3: Backup & Restore Automation Framework – Multi-Platform, Multi-Storage

- Developed a backup and restore automation framework spanning FMC, FTD-managed devices, and cdFMC environments.
- Automated full, scheduled, and cross-environment backup/restore workflows with integrity verification at each stage.
- Integrated SMB, NFS, SSH/SFTP, and Amazon S3 backends with network path validation, bandwidth testing, and endpoint reachability checks.
- Engineered failure-scenario coverage – network timeouts, unreachable endpoints, interrupted transfers, corrupted artifacts – to validate platform resilience.

Project 4: Diagnostic Bundle Generation & Validation Framework

- Designed an automation framework to generate and validate diagnostic bundles across FMC, FTD, and cdFMC with network connectivity diagnostics and endpoint health checks.
- Validated bundle integrity, network state capture, required-data completeness, and support-readiness across multiple platform variants to improve triage consistency.
- Engineered negative-path and degraded-condition scenarios to confirm graceful failure handling.
- Integrated into CI/CD for continuous assurance of diagnostic artifact consistency across releases.

Project 5: Cross-Platform Migration Validation Framework – FMC

- Implemented a cross-platform automation framework validating FMC migration workflows with network path validation, connectivity checks, and endpoint reachability across VMware, AWS, GCP, and physical appliance environments.
- Automated migration execution and post-migration network validation checks – configuration persistence, policy integrity, and operational continuity.
- Validated rollback paths to ensure safe failure handling during migration runs.
- Integrated into CI/CD for continuous cross-platform migration regression coverage.

Technical Skills

Programming: Python (primary), Java, SQL, Shell Scripting

Automation: pyATS, pytest, REST/RPC API Automation, CLI Automation, Test Framework Design

CI/CD: Jenkins, Pipeline Automation, Parallel Execution, Logging & Monitoring

Networking & Security: TCP/IP, DNS, HTTP/S, SSL/TLS, Routing & Switching, Cisco FTD (firewall), FMC & cdFMC, High Availability, CCNA

Cloud & Infrastructure: AWS (EC2, S3), GCP, VMware, on-prem / bare-metal appliances

Storage Protocols: SMB, NFS, SSH/SFTP, Amazon S3

Concepts: Network Troubleshooting, Linux Networking, IP Reachability, Network Topology, Endpoint Validation, Bandwidth Testing, Disaster Recovery, Backup & Restore, Failure Testing, Observability, Runbooks / Documentation

AI Tools: GitHub Copilot, Cursor, Windsurf

Tools: Git, Jira, Postman, Linux, VS Code, PyCharm

Achievements

KHOJ – Agentic Onboarding Assistant

Apr 2024

Winner – Cisco Internal Hackathon (Team Pragya)

Bangalore, India

- Won **1st place** at Cisco Internal Hackathon (2024) for building **KHOJ**, an LLM-powered onboarding assistant deployed as a **Webex bot**.
- Built the conversational backend integrating **Cisco's internal LLM** with **MCP**-based service orchestration to automate access requests, answer org-specific queries, and surface learning resources.
- Delivered end-to-end as the **sole developer** within **72 hours**, owning LLM integration, MCP server setup, and Webex bot deployment.

Certifications

- CCNA – Cisco Certified Network Associate
- Python Programming – Simplilearn
- Command Line in Linux – Coursera
- Microsoft AI Classroom – Microsoft
- Java Programming – Simplilearn

Education

Madanapalle Institute of Technology & Science

Bachelor of Technology in Computer Science (GPA: 7.74 / 10.00)

May 2019 – May 2023

Madanapalle, India